

ROSE MARIA

📞 (+91) 9846825969 | ✉ ro.maria5969@gmail.com | 🔗 rose-maria-911340296 | 🌐 mariaR5

EXPERIENCE

beataboutmusic

January 2026 – Present

Product Engineer – Recommendation Systems

Remote

- Architected a fully functional rule-based music recommendation engine using FastAPI and Pandas, performing deep analysis of implicit user signals (clicks, likes, follows) and content-based filtering for similar artist discovery.
- Devised a cold start handler using genre-based candidate generation, stratified sampling, and round-robin diversification to deliver balanced artist recommendations.
- Developed and maintained full-stack product features using Node.js, Vue.js, MySQL, and Docker, contributing to core platform functionality and deployment workflows.

Crink.app

March 2026 – May 2026

AI Engineer Intern

KSUM, Kochi

- Developed a synthetic conversation generation system using GPT-5.4 to simulate 30 distinct user personas for two LLM-based AI coaches.
- Designed an LLM-as-judge pipeline evaluating responses across 6 quality dimensions against 15 therapist-validated ground truths per conversation.
- Established an automated accuracy measurement system that quantifies coach response quality by computing ground truth pass rates across evaluation runs, enabling iterative model benchmarking.

Unified Mentor

June 2025 – August 2025

Machine Learning Intern

Remote

- Developed and evaluated multiple machine learning solutions spanning regression, classification, and computer vision, applying end-to-end workflows from data preprocessing to model deployment.
- Built and optimized predictive models using XGBoost, Random Forest, and Logistic Regression, performing feature engineering, hyperparameter tuning, and model evaluation with metrics such as MAE, RMSE, R^2 , F1-score, and ROC-AUC.
- Deployed machine learning applications using Flask and Streamlit, integrating trained models with responsive web interfaces for real-time inference.

Actionfi Technologies

May 2024

Data Analytics Intern

UL Cyberpark, Calicut

- Performed exploratory data analysis (EDA) on a 10,000-row bot performance dataset using Pandas and Matplotlib, generating visualizations and insights into automation pipeline efficiency.
- Built a Gemini API-powered AI system for structured data extraction from newspaper clippings, prototyping a research paper digitization workflow.
- Developed an LLM benchmarking framework evaluating GPT-3.5 Turbo, GPT-4, and GPT-4o against expected outputs to automate model selection.

PROJECTS

Cue – Universal Music Scrobbler 🎧

- Architected a cross-platform music tracking ecosystem in Flutter and FastAPI that intercepts Android media notifications to scrobble playback across 16 music apps.
- Designed a multi-modal AI recommendation engine powered by Gemini 2.5 Flash for lyric, genre, production-credit, and sample-based music discovery using Genius and MusicBrainz APIs.

- Integrated JWT-based authentication with email OTP verification, PostgreSQL persistence via SQLAlchemy, and Spotify Web API for metadata validation and cover art retrieval.

Bylaw Buddy – RAG-Based Legal Chatbot 🗣️

- Developed a full-stack RAG pipeline using LangChain, FAISS vector search, and HuggingFace embeddings (all-MiniLM-L6-v2) to answer natural language queries across Indian law PDFs with page-level citations.
- Built PDF highlighting using PyMuPDF and BrightData web-search fallback, deploying a FastAPI + Streamlit application powered by Groq-hosted Llama 3.3-70B.

Clickflow – Automated Flowchart Digitization 🗣️ 🌐

- Designed a 3-tier full-stack platform where Gemini Flash (multimodal) parses hand-drawn flowchart images and natural language prompts into structured JSON node-edge configurations rendered using React-Flow.
- Engineered an asynchronous FastAPI backend using SQLAlchemy, Pydantic, and PostgreSQL (JSONB), exposing REST APIs and a custom graph auto-layout engine for interactive editing.

ASL Detection – Real-Time Sign Language Recognition 🗣️

- Built a real-time American Sign Language recognition system using MediaPipe and OpenCV to detect 28 static hand gestures from webcam input.
- Generated landmark-based datasets from MediaPipe hand tracking and trained a Random Forest classifier for live gesture recognition.
- Integrated the trained model with OpenCV to perform real-time inference and display predicted gestures directly on the video stream.

SKILLS

Languages: Python, JavaScript, SQL, Dart, C++, Java, C

AI/ML: LLMs, RAG, Recommendation Systems, LLM Evaluation, LLM-as-Judge, Prompt Engineering, Synthetic Data Generation, Classification, Regression, Feature Engineering, FAISS, HuggingFace Embeddings

Frameworks & Libraries: FastAPI, LangChain, Pandas, NumPy, Scikit-learn, Flask, Streamlit, Flutter, Vue.js, React, Node.js

Databases & Cloud: PostgreSQL, MySQL, SQLite, Azure

Tools & APIs: Docker, Git, GitHub, REST APIs, OpenAI API, Gemini API, Spotify API

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems

ACHIEVEMENTS

Hackathon (TinkHerHack), by TinkerHub

Kerala's largest women-only hackathon; recognized as the top team at the venue and among the top 300 teams statewide (2-member team). Built Bylaw Buddy, a RAG-based legal chatbot, as the sole backend developer, implementing the RAG pipeline, FastAPI server, LangChain orchestration, and Groq-hosted LLM integration.

EDUCATION

Cochin University of Science and Technology

2023–2027

B.Tech in Computer Science & Engineering

CGPA: 9.52 (S5)

G V H S S for Girls, Nadakkavu

2021–2023

Higher Secondary – Science with Computer Science

99.8%